Blast Radius of Pollution Attacks in Quantum Network Coding: A Circuit-Level Study of Fidelity Collapse and Error Propagation

A circuit-level Qiskit study showing that when classical correction bits are mixed at a network bottleneck, corrupting a single packet collapses state fidelity at every destination that depends on it, while the same attack on a dedicated channel harms only one.

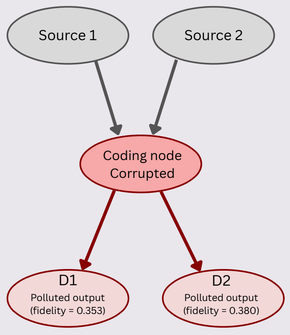
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Introduction

Quantum network coding (QNC) boosts throughput by letting intermediate nodes combine several sources' quantum information, but this means one corrupted node can pollute every destination downstream. Defending against it is hard: no-cloning prevents checking a state against a backup, and teleportation relies on a classical correction channel that an attacker can tamper with without touching any qubit. This study compares six pollution-resistant QNC schemes (2016-2024) on five axes, identifies an open gap in topology and threat coverage, and uses Qiskit simulations to show that corrupting one classical packet that mixes two flows' correction bits degrades both destinations, while the same attack on a dedicated channel harms only one.

Problem Statement

While quantum network coding (QNC) increases throughput via joint multi-source operations, a single compromised node can corrupt all downstream destinations in a multi-destination pollution attack. Defense is hindered by the no-cloning theorem, which prevents state verification via backup copies and the dependence on an auxiliary classical correction channel that adds a vulnerable non-quantum attack surface. Consequently, this study evaluates how existing protocols maintain throughput while examining their trade-offs in trust, resource overhead, and topological generality, alongside the conditions under which classical correction-packet corruption does, or does not, propagate across multiple destinations.



Example trial

Evolution of Pollution-Resistant QNC Schemes

Scheme	Trust assumption	Classical channel	Detection	Threat Scope	Topology Scope
Homomorphic signature	Trusted nodes required	Required	Detect + locate	Pollution only	Butterfly only
Continuous variable QNC	None	Required	Detect	Pollution only	Butterfly only
Single-shot general-network	None	Required (free/public)	Detect	Pollution + limited leakage	General k-pair
Channel-Free Secret-Randomness	None	Eliminated	Detect	Pollution only	Butterfly only
QFHE / d-Dimensional	None	Required	Detect, collusion-resilient	Pollution + leakage	Butterfly only
Complete Security	None	Required	Detect	Pollution + leakage (unified)	Butterfly only

Table 1: Evolution of Pollution-Resistant QNC Schemes

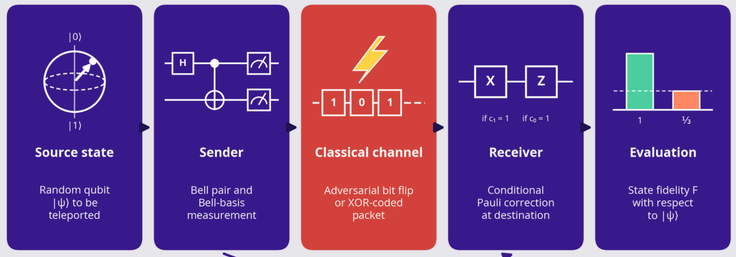
References

[1] M. Hayashi, K. Iwama, H. Nishimura, R. Raymond, and S. Yamashita, "Quantum network coding," in Proc. STACS 2007, Lecture Notes in Computer Science, vol. 4393, pp. 610-621, 2007.

[2] T. Shang, Z. Pei, X. J. Zhao, and J. W. Liu, "Quantum network coding against pollution attacks," IEEE Communications Letters, vol. 20, no. 7, pp. 1369-1372, Jul. 2016.

[3] D. X. Charles, K. Jain, and K. Lauter, "Signatures for network coding," invited paper, CISS 2006.

Methodology



Evaluation Protocol

I. Pollution model
Correction-bit tampering in a single teleportation link (N = 300)

II. Authentication
Verification with n-bit tags, n = 1 to 8 (N = 500 per tag length)

III. Error propagation
XOR-coded bottleneck, m = a ⊕ b (N = 500 paired trials)

Attack model
The adversary flips classical correction bits in transit. No qubit is touched, and the flips are exact rather than probabilistic noise.

Metric
State fidelity F = $|\langle\psi|\varphi\rangle|^2$ between the target and received states, over random input states. Any single Pauli error gives a mean of 1/3, which is the baseline shown in the diagram.

Bottleneck
Two teleportation links share one packet m = a ⊕ b. Each destination recovers its own bits using side information sent directly by the other source. These trials are paired and seeded.

Scope
Noise-free statevector simulation, a toy n-bit authentication tag, and classical XOR mixing rather than a joint quantum operation at the coding node.

Empirical Fidelity Analysis

This section reports circuit-level Qiskit results in three parts: single-link pollution, detection, and error propagation through a mixed-correction bottleneck.

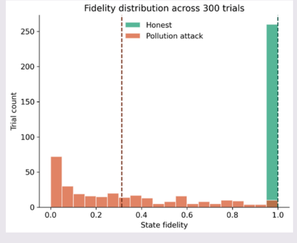
Single-Link Pollution

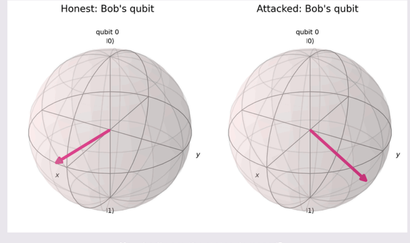
Over 300 randomized trials, honest teleportation gave a fidelity of 1.000 with zero variance. With the classical correction bits flipped in transit, mean fidelity fell to 0.313 (SD 0.285) and spread across the whole range from near 0 to near 1.

This matches theory. A flipped correction bit leaves a Pauli error, whose fidelity is the squared Bloch component along the error axis, averaging 1/3 over random states. Fig. 2 shows one case: the attacked Bloch vector is the honest vector rotated by π about the z-axis, with no qubit touched.

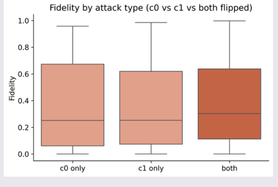
Corrupted bit	Mean fidelity	Trials
c0 only	0.343	300
c1 only	0.300	300
c0 and c1	0.312	300

Table 2: Mean fidelity by corrupted bit

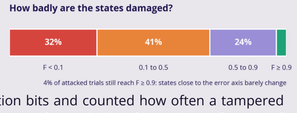




Which bit is corrupted does not matter (Table 2, Fig. 3). All three means sit near 1/3, as expected for any Pauli error.



About 72% of attacked trials fell below F = 0.5, and only 4% stayed above 0.9. The damage depends on how far the input state lies from the error axis.



Detection

We added an n-bit authentication tag over the classical correction bits and counted how often a tampered packet was rejected. With a 4-bit tag, 94.33% of attacks were detected over 300 trials, against a theoretical 93.75% (binomial SE about 1.4%). Across tag lengths, simulated detection follows the $1 - 2^{-n}$ curve (Table 3, Fig. 4).

Tag bits	Theoretical	Simulated
1	0.500	0.488
2	0.750	0.772
4	0.9375	0.936
8	0.9961	1.000

Table 3. Detection rate vs tag length

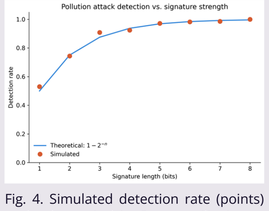
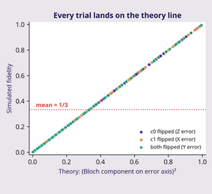


Fig. 4. Simulated detection rate (points) vs the theoretical $1 - 2^{-n}$ curve (line).

Each extra bit halves the chance that a forged packet passes. This confirms the expected forgery probability of an n-bit tag. It does not validate any specific signature scheme, because the tag here is a toy stand-in (Section 5.5).



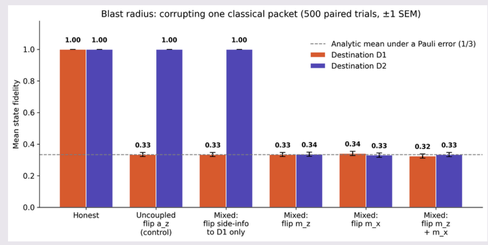
Blast Radius: Dedicated vs XOR-Mixed Channels

To test how corruption spreads, we simulated a butterfly-style bottleneck in which two teleportation links' correction bits are combined into one packet, m = a ⊕ b. Each destination recovers its own bits using side information sent directly by the other source. We ran 500 paired trials per scenario, flipping one bit and measuring fidelity at both destinations (Table 4, Fig. 5).

Scenario	F(D1)	F(D2)	Destinations hit
Honest	1.000	1.000	0
Dedicated channel: flip a_z (control)	0.344 ± 0.014	1.000	1
XOR-mixed: flip m_z	0.362 ± 0.014	0.346 ± 0.013	2
XOR-mixed: flip m_x	0.338 ± 0.014	0.342 ± 0.013	2
XOR-mixed: flip m_z and m_x	0.323 ± 0.013	0.328 ± 0.013	2
XOR-mixed: flip side-info to D1 only	0.317 ± 0.014	1.000	1

Table 4. Blast radius: mean fidelity at D1 and D2 when one classical bit is corrupted (±1 SEM, 500 paired trials).

Corrupting one bit of the mixed packet damages both destinations (F ≈ 0.33 each), while the same flip on a dedicated channel damages only D1. Flipping only the direct side-information link also harms one destination, because only the mixed packet couples the two flows.



Flip one bit

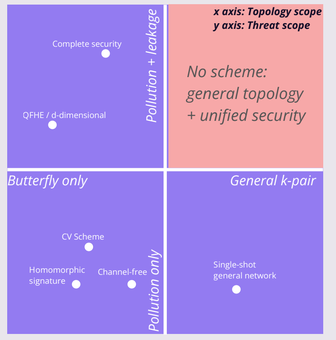
Dedicated: flip a
✗ D1 ✓ D2

XOR-mixed: flip m
✗ D1 ✗ D2

Interpreting the Results

Trend Across Schemes

Each scheme removes roughly one assumption left open by the one before: trusted nodes, then the classical channel, then collusion, then leakage. The continuous-variable scheme shows the vulnerability is independent of encoding. Guarantees have strengthened over time, but no scheme combines every improvement.



Identified Gap

Table 1 shows that unified pollution and leakage resistance has only been demonstrated on the butterfly network, while general k-pair topologies are covered only for narrower threat models. No surveyed scheme shows unified security on a general topology.

1.000 → 0.32
mean fidelity: honest vs bit-flip attack (analytic mean for a Pauli error: 1/3)

≈ 94%
attacks detected with a 4-bit tag (theory: $1 - 2^{-n} = 93.75\%$)

2 of 2 vs 1 of 2
destinations damaged: mixed packet vs dedicated channel

Discussion

A flipped correction bit drops fidelity to about 1/3 whichever bit is flipped. What matters is how far the damage spreads: corrupting one XOR-mixed packet harmed both destinations, while the same flip on a dedicated channel harmed only one. Mixing correction data, not merely sharing a channel, creates the blast radius.

Conclusion

Mixing is what gives a quantum network coding node its blast radius, and a simple authentication tag can detect the tampering. These are ideal-simulator results with classical XOR mixing and a toy tag, so a full butterfly network, a proven signature, and hardware tests remain future work.